Telco_Churn_EDA

April 18, 2025

B"H

1 Telecom Customer Churn - Exploratory Data Analysis

1.1 DSC-630 Milestone 3 - Preliminary Analysis

1.1.1 David Koyrakh

This notebook contains the preliminary exploratory data analysis for the telecom customer churn prediction project. The goal is to assess whether the data can effectively answer the research questions and identify any adjustments needed to my approach.

1.2 1. Loading the Dataset and Initial Inspection

In this section, we'll load the IBM Telco Customer Churn dataset and perform an initial inspection to understand its structure, variable types, and basic characteristics. This first overview will help us identify potential data quality issues and get a sense of the dataset's completeness.

[65]:
```
# Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

# Set display options
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 1000)

# Set visualization style
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette('Set2')
plt.rcParams['figure.figsize'] = (12, 8)
plt.rcParams['font.size'] = 12
```

[66]:
```
# Load the dataset
df = pd.read_csv('Telco-Customer-Churn.csv')

# Display basic information
```

```python
print(f"Dataset shape: {df.shape}")
df.head()
```

```
Dataset shape: (7043, 21)
```

[66]:
```
   customerID  gender  SeniorCitizen Partner Dependents  tenure PhoneService
MultipleLines InternetService OnlineSecurity OnlineBackup DeviceProtection
TechSupport StreamingTV StreamingMovies        Contract PaperlessBilling
PaymentMethod  MonthlyCharges TotalCharges Churn
0  7590-VHVEG  Female              0     Yes         No       1           No  No
phone service             DSL             No           Yes                 No
No          No              No  Month-to-month             Yes
Electronic check           29.85        29.85    No
1  5575-GNVDE    Male              0      No         No      34           Yes
No             DSL             Yes           No               Yes          No
No              No        One year              No              Mailed check
56.95       1889.5    No
2  3668-QPYBK    Male              0      No         No       2           Yes
No             DSL             Yes          Yes                No          No
No              No  Month-to-month             Yes              Mailed check
53.85       108.15   Yes
3  7795-CFOCW    Male              0      No         No      45           No  No
phone service             DSL            Yes            No               Yes
Yes          No              No        One year               No  Bank transfer
(automatic)           42.30      1840.75    No
4  9237-HQITU  Female              0      No         No       2           Yes
No     Fiber optic              No           No                No          No
No              No  Month-to-month             Yes          Electronic check
70.70       151.65   Yes
```

The dataset contains 7,043 customer records with 21 features. Looking at the first few rows, we can see that the dataset includes a good mix of demographic information (gender, senior citizen status, partner status), service subscriptions (phone, internet, streaming services), contract details, and billing information. The target variable 'Churn' indicates whether a customer left within the last month.

[67]:
```python
# Check data types and missing values
print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())
```

```
Data Types:
customerID          object
gender              object
SeniorCitizen        int64
Partner             object
```

```
Dependents           object
tenure                int64
PhoneService         object
MultipleLines        object
InternetService      object
OnlineSecurity       object
OnlineBackup         object
DeviceProtection     object
TechSupport          object
StreamingTV          object
StreamingMovies      object
Contract             object
PaperlessBilling     object
PaymentMethod        object
MonthlyCharges      float64
TotalCharges         object
Churn                object
dtype: object

Missing Values:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

Examining the data types, I notice that most features are objects (categorical variables), with a few numerics like 'SeniorCitizen', 'tenure', and 'MonthlyCharges'. Interestingly, 'TotalCharges' is stored as an object type rather than numeric, which will need to be converted for analysis.

The dataset appears to be complete with no missing values, which is excellent for the analysis as I won't need to handle imputation for missing data.

[68]:
```
# Summary statistics for numerical columns
df.describe()
```

[68]:
```
       SeniorCitizen       tenure  MonthlyCharges
count    7043.000000  7043.000000     7043.000000
mean        0.162147    32.371149       64.761692
std         0.368612    24.559481       30.090047
min         0.000000     0.000000       18.250000
25%         0.000000     9.000000       35.500000
50%         0.000000    29.000000       70.350000
75%         0.000000    55.000000       89.850000
max         1.000000    72.000000      118.750000
```

Looking at the numerical variables:

- 'SeniorCitizen' is a binary variable (0 or 1) with only about 16% of customers being seniors
- 'tenure' ranges from 0 to 72 months, with a mean of about 32 months - indicating significant variability in customer longevity
- 'MonthlyCharges' has a wide range ($18.25 to $118.75) with a mean of $64.76

The distribution of tenure is particularly interesting, as it will likely be a strong predictor of churn behavior.

[69]:
```
# Summary for categorical columns
df.describe(include=['object'])
```

[69]:
```
        customerID gender Partner Dependents PhoneService MultipleLines
InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport
StreamingTV StreamingMovies        Contract PaperlessBilling     PaymentMethod
TotalCharges Churn
count         7043   7043    7043       7043         7043          7043
7043           7043         7043             7043        7043        7043
7043            7043             7043             7043          7043  7043
unique        7043      2       2          2            2             3
3              3            3                3           3           3
3               3                2                4         6531     2
top     7590-VHVEG   Male      No         No          Yes            No
Fiber optic             No           No                No          No          No
No  Month-to-month              Yes  Electronic check              No
freq             1   3555    3641       4933         6361          3390
3096           3498         3088             3095        3473        2810
2785            3875             4171             2365            11  5174
```

The categorical variable summary provides useful insights:

- Gender is somewhat balanced (slightly more males than females)
- Most customers don't have partners or dependents
- Fiber optic is the most common internet service
- Most customers are on month-to-month contracts
- About 73% of customers haven't churned (5,174 'No' vs. 1,869 'Yes')

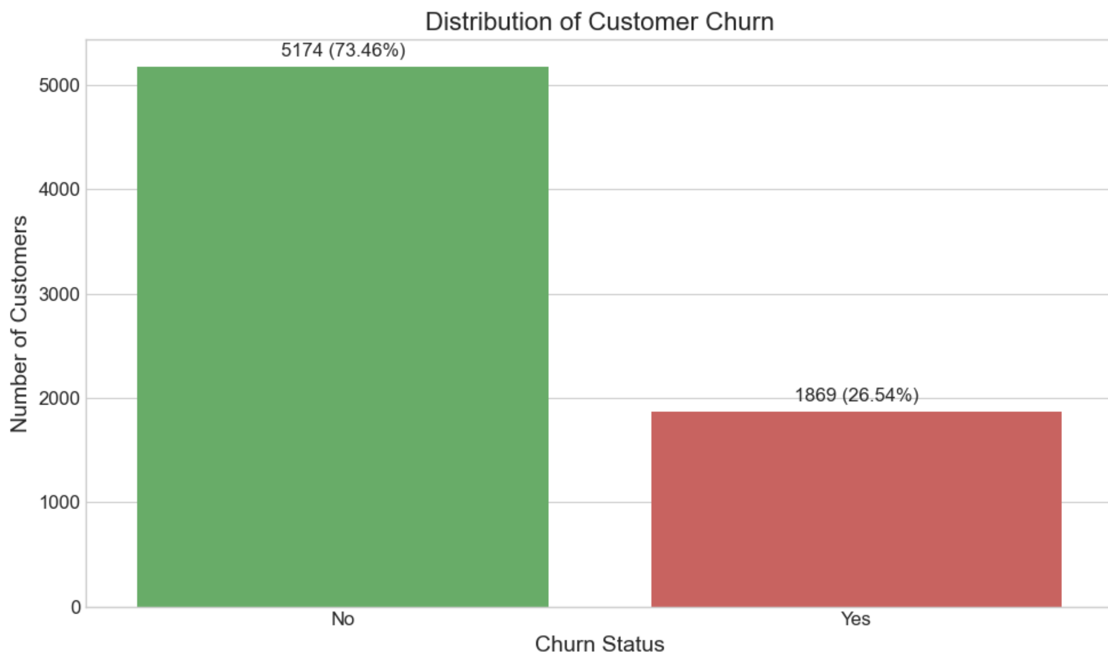
These initial findings suggest areas to explore further, especially the relationship between contract type and churn rate.

1.3 2. Target Variable Analysis: Churn

Now we'll examine the target variable - customer churn. Understanding the distribution of churned versus retained customers is crucial for assessing class imbalance, which will inform my modeling strategy and evaluation metrics.

[70]:
```python
# Distribution of the target variable
churn_counts = df['Churn'].value_counts()
churn_pct = round(df['Churn'].value_counts(normalize=True) * 100, 2)

plt.figure(figsize=(10, 6))
ax = sns.countplot(x='Churn', data=df, palette=['#5cb85c', '#d9534f'])

# Add count and percentage labels
for i, (count, pct) in enumerate(zip(churn_counts, churn_pct)):
    ax.text(i, count + 100, f'{count} ({pct}%)', ha='center', fontsize=12)

plt.title('Distribution of Customer Churn', fontsize=16)
plt.ylabel('Number of Customers', fontsize=14)
plt.xlabel('Churn Status', fontsize=14)
plt.tight_layout()
plt.show()
```

```
C:\Users\David\AppData\Local\Temp\ipykernel_6772\1767592563.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  ax = sns.countplot(x='Churn', data=df, palette=['#5cb85c', '#d9534f'])
```



The chart shows that approximately 26.54% of customers in the dataset have churned, while 73.46% remain with the company. This represents a moderate class imbalance that I'll need to consider during modeling.

The imbalance is not extreme (it's not a rare event classification problem), but it's significant enough that I should consider techniques such as: * Using appropriate evaluation metrics (recall, F1-score, AUC-ROC) instead of just accuracy * Potentially applying class weights or sampling techniques like SMOTE during modeling * Ensuring the train-test splits maintain this class distribution

Having about 1,869 churned customers in the dataset should provide sufficient examples for the model to learn patterns associated with churn behavior.

Let's examine the class imbalance in the target variable. This is important for choosing appropriate evaluation metrics and potentially implementing techniques to address imbalance (like SMOTE or class weighting).

1.4 3. Feature Analysis

1.4.1 3.1 Numerical Features

Let's examine the numerical features in the dataset to understand their distributions and relationships with customer churn. This will help us identify potential predictors and necessary transformations before modeling.

[71]:
```
# Identify numerical features (excluding SeniorCitizen which is actually␣
↪categorical)
numeric_features = ['tenure', 'MonthlyCharges', 'TotalCharges']
```

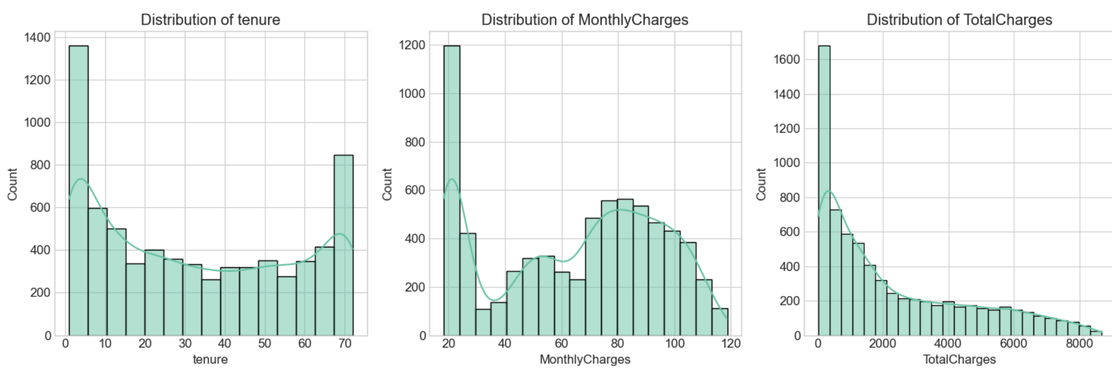
```python
# Check the data type of TotalCharges
print(f"TotalCharges data type: {df['TotalCharges'].dtype}")

# Convert TotalCharges to numeric
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')

# Check if there are any null values after conversion
print(f"Number of null values in TotalCharges after conversion:␣
↪{df['TotalCharges'].isnull().sum()}")

# Drop or fill the rows with null values in TotalCharges
df = df.dropna()
print(f"Dataset shape after removing rows with null values: {df.shape}")
```

```
TotalCharges data type: object
Number of null values in TotalCharges after conversion: 11
Dataset shape after removing rows with null values: (7032, 21)
```

We've successfully converted the 'TotalCharges' column to numeric, identifying and handling a few rows (11) with non-numeric values. These likely represent new customers with no accumulated charges yet. For simplicity, we've removed these rows, but in a more comprehensive analysis, we might consider imputing these values or treating them as a separate category.

Now let's visualize the distributions of the numerical features to better understand their characteristics.

[72]:
```python
# Analyzing distributions of numerical features with histograms
plt.figure(figsize=(15, 5))
for i, feature in enumerate(numeric_features):
    plt.subplot(1, 3, i+1)
    sns.histplot(df[feature], kde=True)
    plt.title(f'Distribution of {feature}')
plt.tight_layout()
plt.show()
```

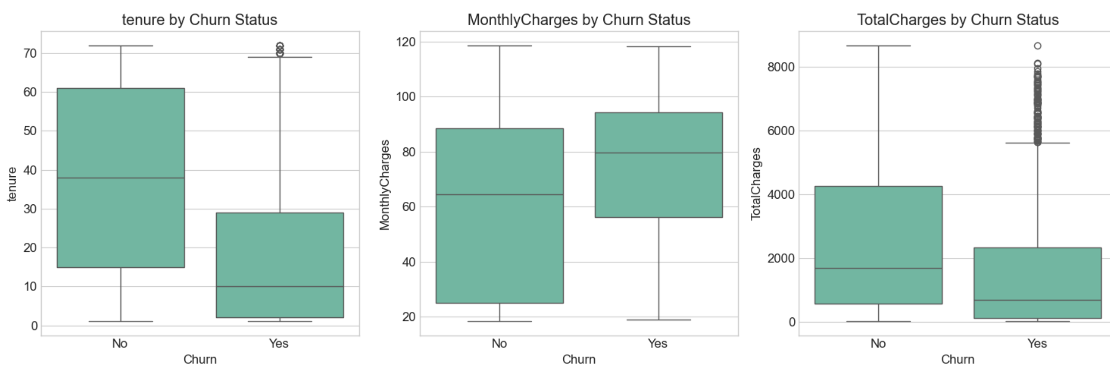


The histograms reveal interesting patterns in the numerical features:

- **Tenure**: Shows a bimodal distribution with peaks at both low and high tenure values. This suggests two main customer groups: relatively new customers and long-term loyal customers. The higher frequency at low tenure may indicate a higher churn rate among newer customers.
- **MonthlyCharges**: Displays a multimodal distribution with several peaks, likely corresponding to different service tiers or package combinations. This suggests that customers cluster around certain price points based on the services they subscribe to.
- **TotalCharges**: Has a right-skewed distribution, with many customers having lower total charges and fewer having very high total charges. This is expected as total charges accumulate over time and are strongly correlated with tenure.

Let's now examine how these numerical features relate to customer churn using box plots.

[73]:
```
# Box plots of numerical features by churn status
plt.figure(figsize=(15, 5))
for i, feature in enumerate(numeric_features):
    plt.subplot(1, 3, i+1)
    sns.boxplot(x='Churn', y=feature, data=df)
    plt.title(f'{feature} by Churn Status')
plt.tight_layout()
plt.show()
```

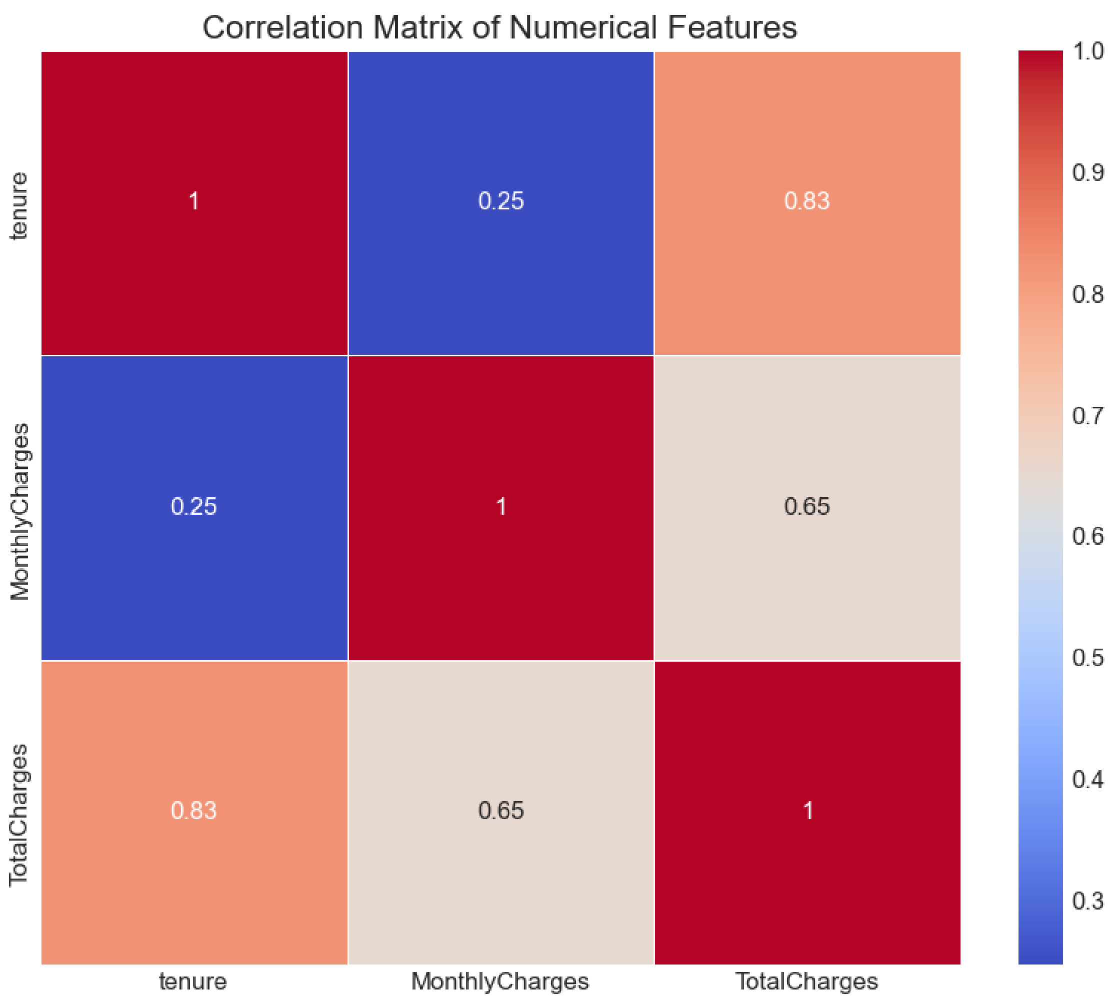


The box plots reveal clear relationships between our numerical features and customer churn:

- **Tenure**: Customers who churn tend to have significantly lower tenure (median around 10 months) compared to those who stay (median around 40 months). This is one of the strongest predictors in the dataset and confirms that newer customers are at higher risk of churning.
- **MonthlyCharges**: Interestingly, customers who churn tend to have higher monthly charges than those who stay. This suggests that customers paying premium prices might be more sensitive to service quality or more likely to shop around for better deals.
- **TotalCharges**: Customers who stay have higher total charges, which is expected given the strong correlation with tenure. This reflects the cumulative value of long-term customers to the business.

Let's also check the correlation between the numerical features to understand their relationships.

[74]:
```python
# Correlation matrix
plt.figure(figsize=(10, 8))
correlation_matrix = df[numeric_features].corr()
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', linewidths=0.5)
plt.title('Correlation Matrix of Numerical Features', fontsize=16)
plt.show()
```



The correlation matrix shows a strong positive correlation (0.83) between 'tenure' and 'TotalCharges', which makes intuitive sense: the longer a customer stays, the more they pay in total.

There's also a moderate correlation (0.65) between 'MonthlyCharges' and 'TotalCharges', which is expected since total charges are influenced by both how much a customer pays monthly and how long they've been a customer.

Interestingly, there's little correlation (0.25) between 'tenure' and 'MonthlyCharges', suggesting that customer longevity doesn't strongly relate to how much they pay monthly. This independent relationship makes both variables potentially useful as separate predictors in our model.

1.4.2 3.2 Categorical Features

Now let's analyze the categorical features in the dataset. These features represent customer demographics and service choices, which could significantly influence churn behavior. We'll visualize the distribution of each categorical variable and its relationship with customer churn.

[75]:
```
# Identify categorical features
categorical_features = [col for col in df.columns if df[col].dtype == 'object'␣
↪or col == 'SeniorCitizen']
categorical_features.remove('Churn')  # Remove target variable from the list
print(f"Categorical features: {categorical_features}")
print(f"Number of categorical features: {len(categorical_features)}")
```

```
Categorical features: ['customerID', 'gender', 'SeniorCitizen', 'Partner',
'Dependents', 'PhoneService', 'MultipleLines', 'InternetService',
'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
'PaymentMethod']
Number of categorical features: 17
```

We've identified the categorical features in the dataset, which include demographic information (gender, senior citizen status, partner, dependents) and service-related choices (phone services, internet services, payment methods, etc.).

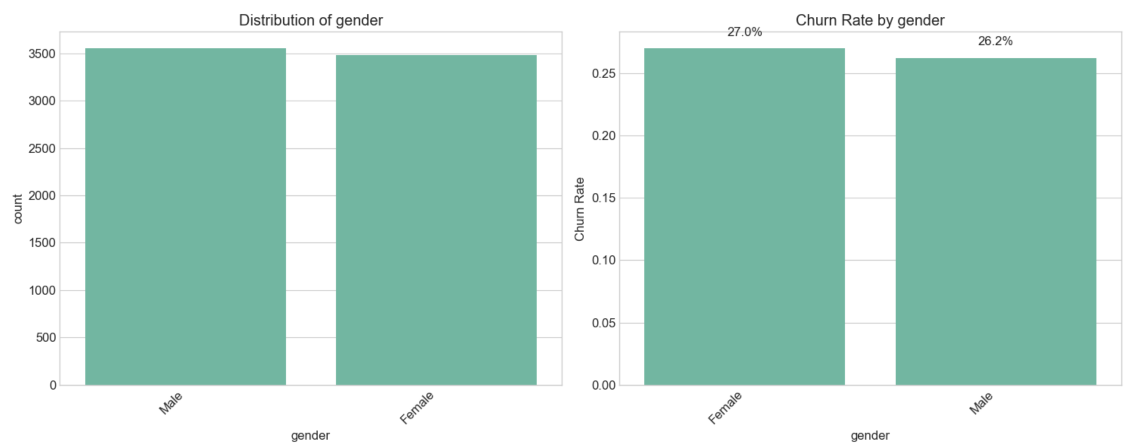
Let's create a function to systematically visualize the distribution of each categorical feature and its relationship with churn.

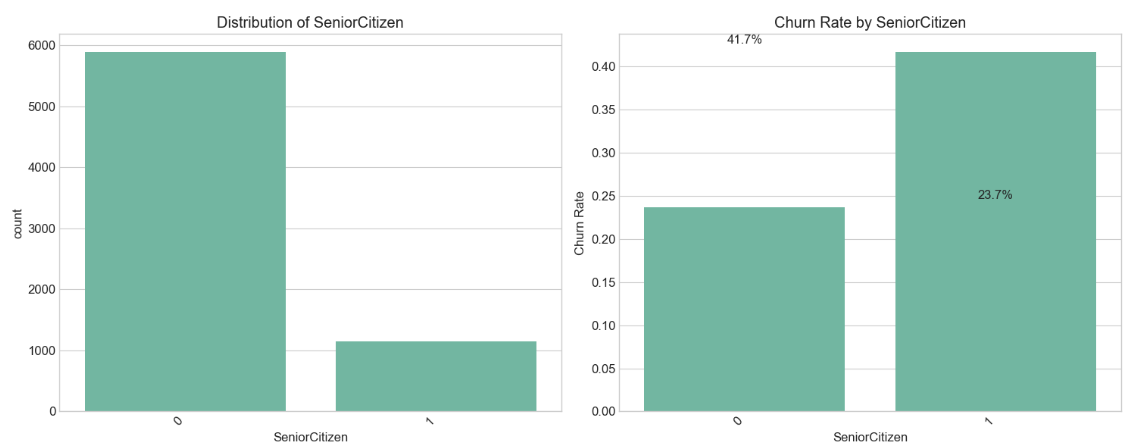
[76]:
```
def plot_categorical_feature(df, feature):
    plt.figure(figsize=(15, 6))

    # Distribution of the feature
    plt.subplot(1, 2, 1)
    sns.countplot(x=feature, data=df, order=df[feature].value_counts().index)
    plt.title(f'Distribution of {feature}')
    plt.xticks(rotation=45, ha='right')

    # Relationship with churn
    plt.subplot(1, 2, 2)
    churn_rates = df.groupby(feature)['Churn'].value_counts(normalize=True).
↪unstack()
    if 'Yes' in churn_rates.columns:
        churn_rates = churn_rates['Yes'].sort_values(ascending=False)
        sns.barplot(x=churn_rates.index, y=churn_rates.values)
        plt.title(f'Churn Rate by {feature}')
        plt.ylabel('Churn Rate')
        plt.xticks(rotation=45, ha='right')

        # Add percentage labels
        for i, rate in enumerate(churn_rates):
```

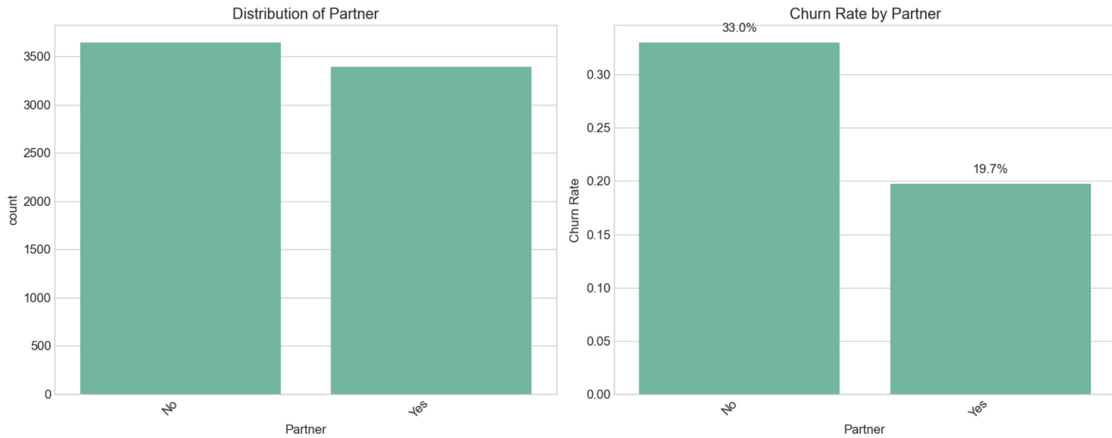
```
            plt.text(i, rate+0.01, f"{rate:.1%}", ha='center')

    plt.tight_layout()
    plt.show()
```

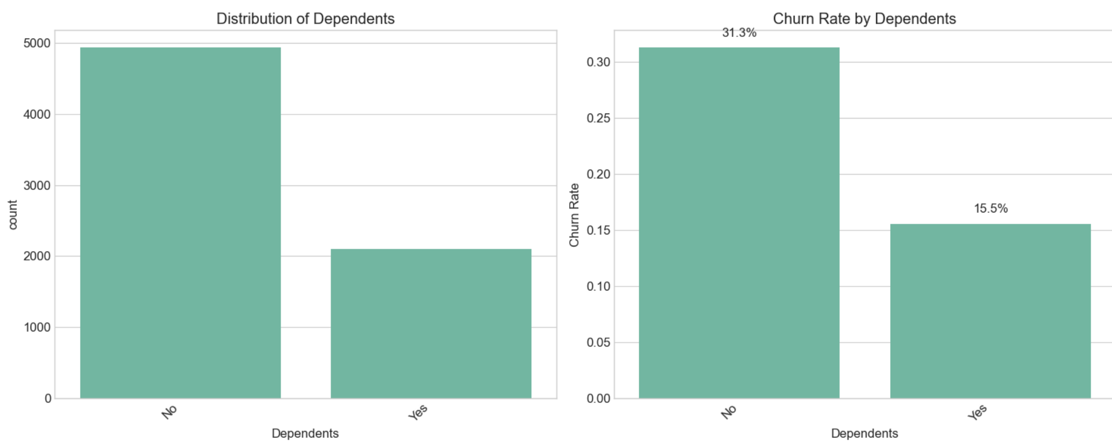
With the visualization function ready, let's analyze key categorical features to understand their impact on customer churn. We'll focus on a few important features first.

[77]:
```
# Analyze demographic features
demographic_features = ['gender', 'SeniorCitizen', 'Partner', 'Dependents']
for feature in demographic_features:
    plot_categorical_feature(df, feature)
```







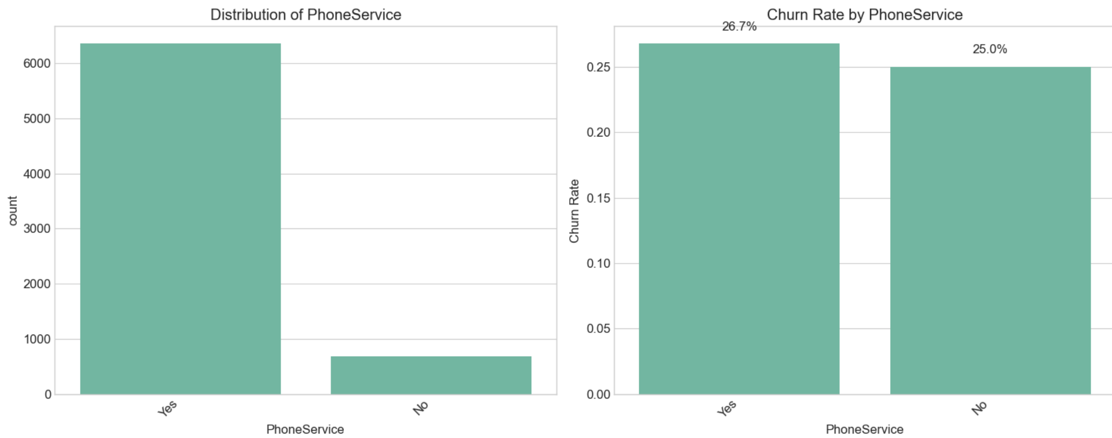


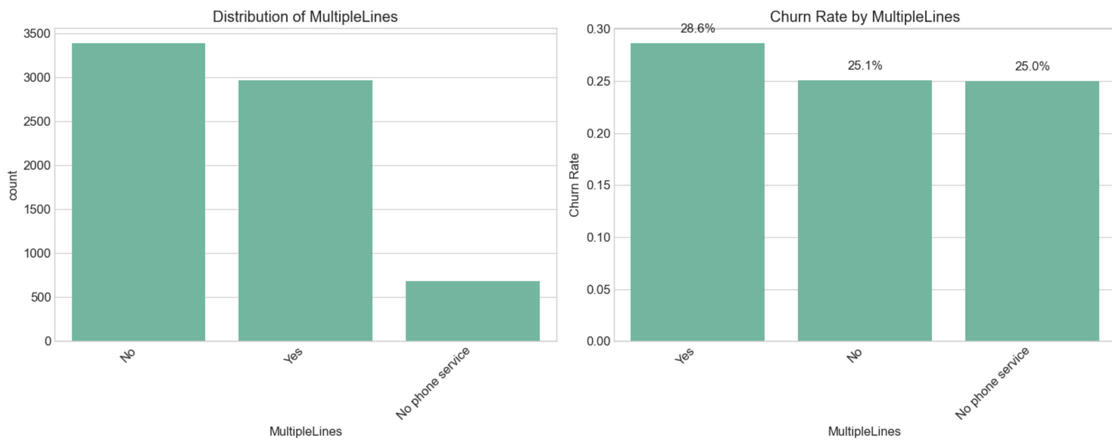
The demographic analysis reveals several interesting patterns:

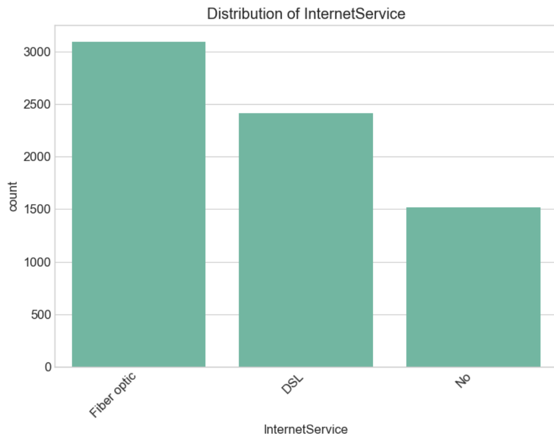
- **Gender**: The customer base is almost evenly split between males and females, with no significant difference in churn rates between genders. This suggests gender is likely not a strong predictor of churn.
- **SeniorCitizen**: While senior citizens represent only about 16% of the customer base, they have a notably higher churn rate (around 42%) compared to non-seniors (about 24%). This indicates that senior citizens may have different needs or face more challenges with the company's services.
- **Partner**: Customers with partners make up approximately half of the customer base and have a lower churn rate (around 20%) compared to those without partners (about 33%). This suggests that having a partner might be associated with more stable service relationships.
- **Dependents**: Customers with dependents represent about 30% of the customer base and have a significantly lower churn rate (around 16%) compared to those without dependents (about 31%). This indicates that family responsibilities might make customers less likely to change service providers.

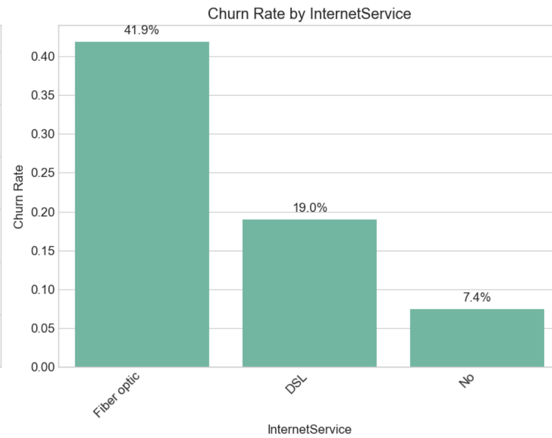
Now, let's analyze the service-related features to understand how different service choices affect churn.

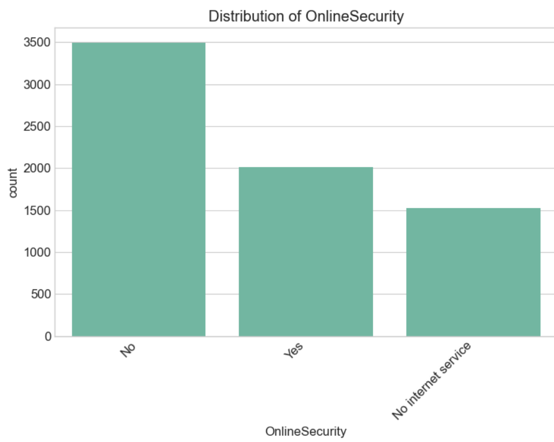
[78]:
```
# Analyze service-related features
service_features = ['PhoneService', 'MultipleLines', 'InternetService',␣
↪'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',␣
↪'StreamingTV', 'StreamingMovies']
for feature in service_features:
    plot_categorical_feature(df, feature)
```

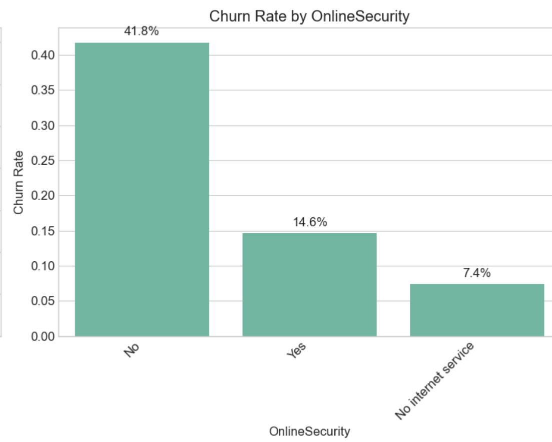


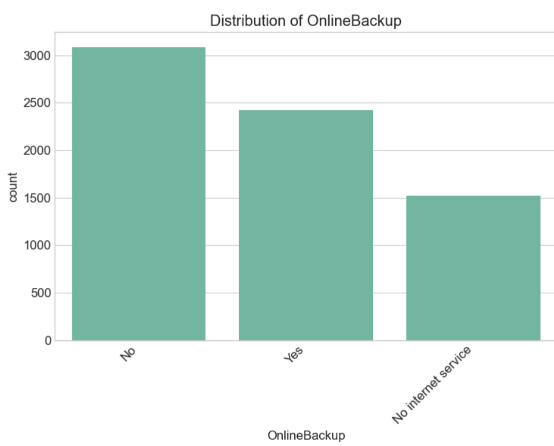


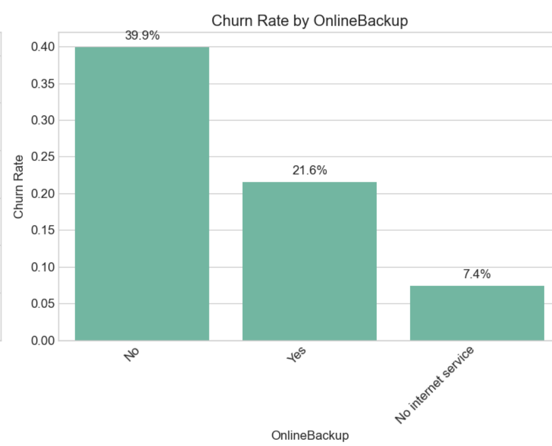

Distribution of InternetService
count
0
500
1000
1500
2000
2500
3000
Fiber optic
DSL
No
InternetService

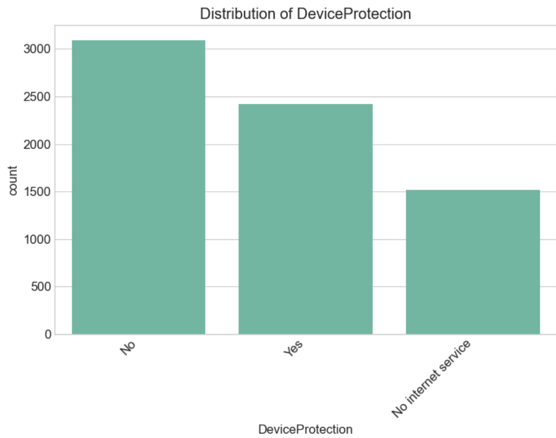

Churn Rate by InternetService
Churn Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
41.9%
19.0%
7.4%
Fiber optic
DSL
No
InternetService

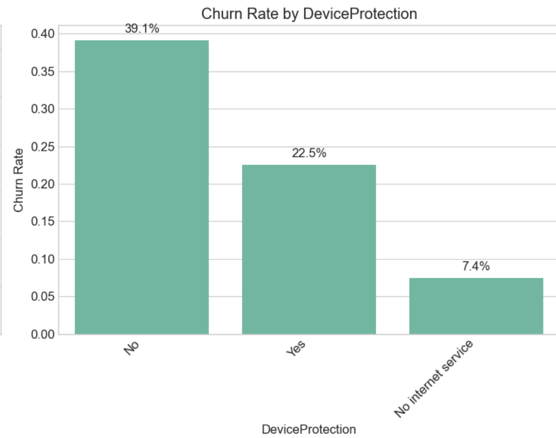

Distribution of OnlineSecurity
count
0
500
1000
1500
2000
2500
3000
3500
No
Yes
No internet service
OnlineSecurity

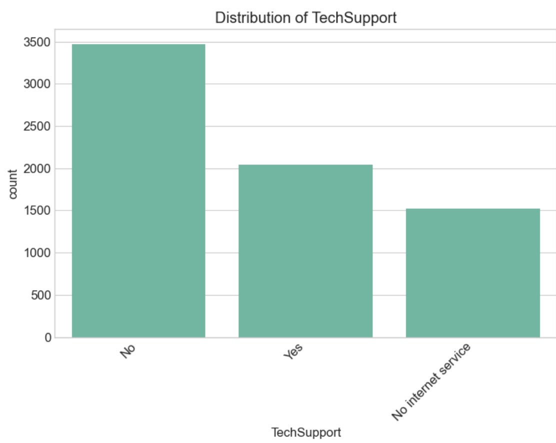

Churn Rate by OnlineSecurity
Churn Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
41.8%
14.6%
7.4%
No
Yes
No internet service
OnlineSecurity

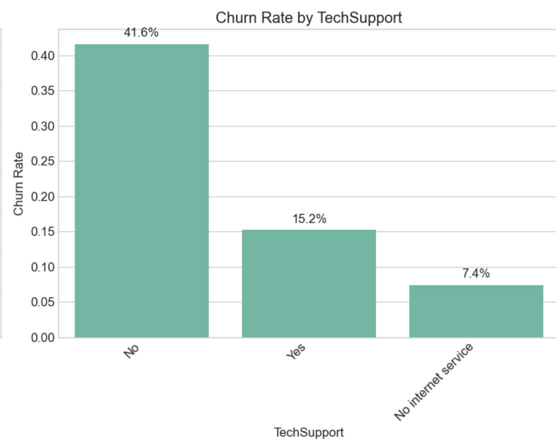

Distribution of OnlineBackup
count
0
500
1000
1500
2000
2500
3000
No
Yes
No internet service
OnlineBackup

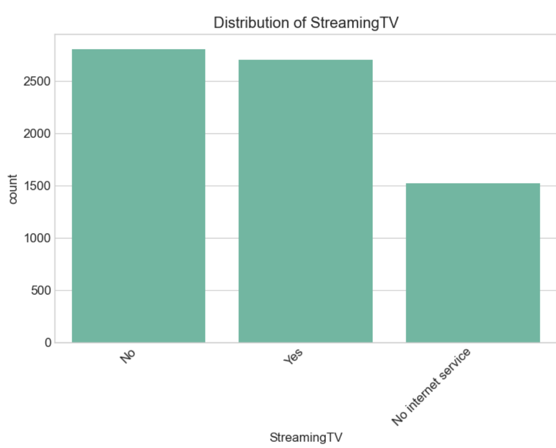

Churn Rate by OnlineBackup
Churn Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
39.9%
21.6%
7.4%
No
Yes
No internet service
OnlineBackup

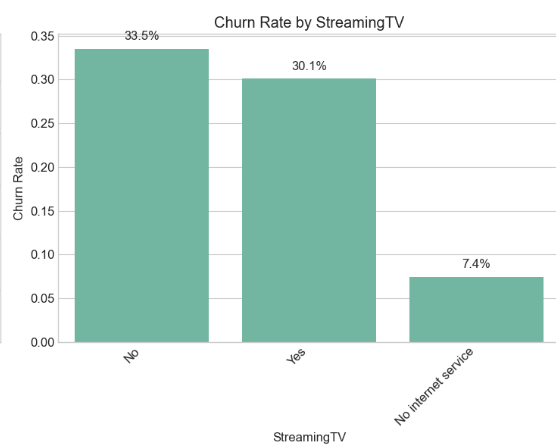

Distribution of DeviceProtection
count
0
500
1000
1500
2000
2500
3000
No
Yes
No internet service
DeviceProtection

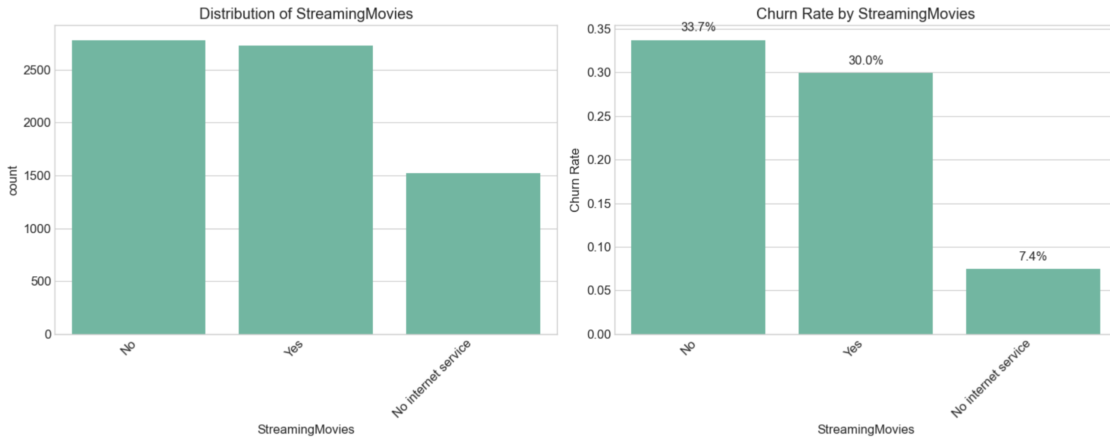

Churn Rate by DeviceProtection
Churn Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
39.1%
22.5%
7.4%
No
Yes
No internet service
DeviceProtection


Distribution of TechSupport
count
0
500
1000
1500
2000
2500
3000
3500
No
Yes
No internet service
TechSupport


Churn Rate by TechSupport
Churn Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
41.6%
15.2%
7.4%
No
Yes
No internet service
TechSupport


Distribution of StreamingTV
count
0
500
1000
1500
2000
2500
No
Yes
No internet service
StreamingTV


Churn Rate by StreamingTV
Churn Rate
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
33.5%
30.1%
7.4%
No
Yes
No internet service
StreamingTV

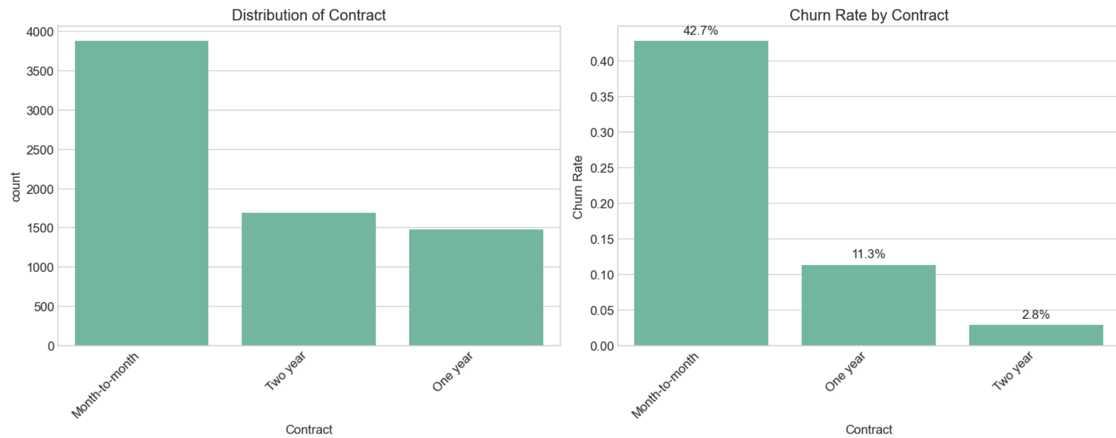


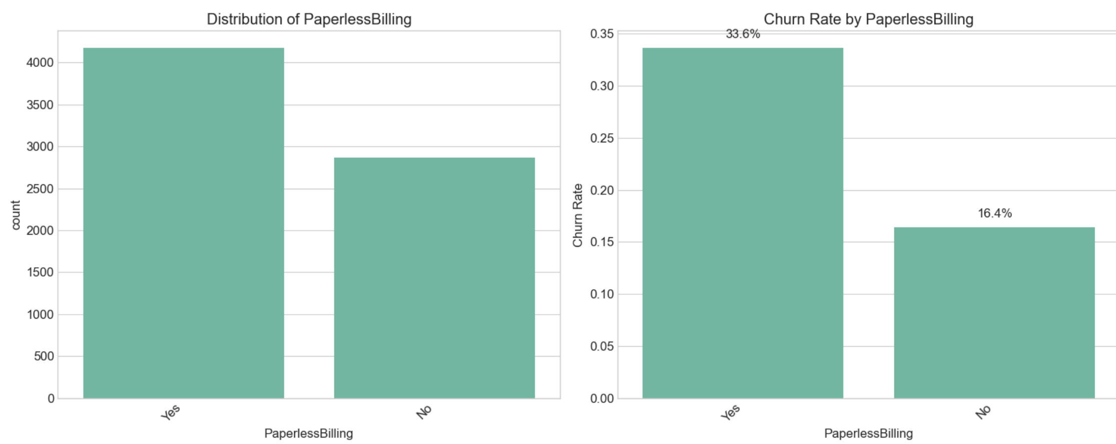
The service-related features analysis highlights several key insights:

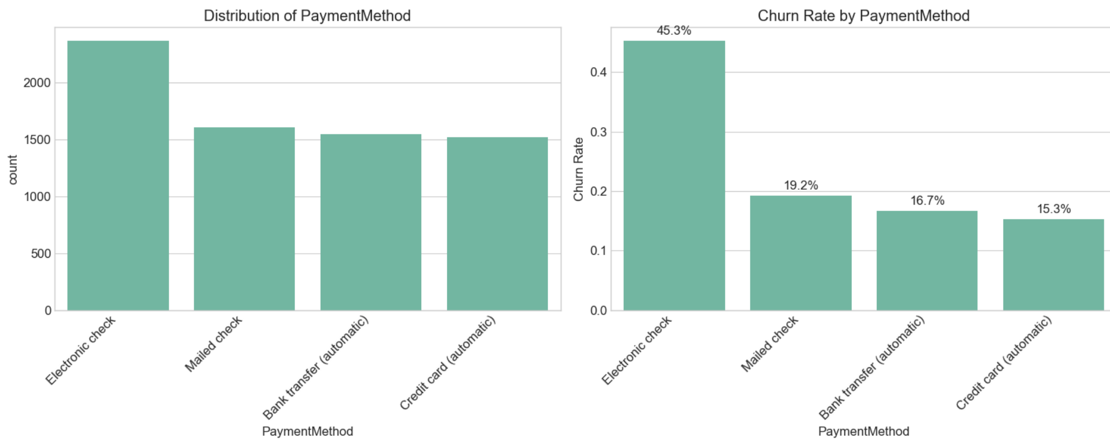
- **PhoneService**: The vast majority (about 90%) of customers have phone service, but this alone doesn't strongly differentiate churn rates.
- **MultipleLines**: Among customers with phone service, those with multiple lines have slightly higher churn rates than those with single lines.
- **InternetService**: This is a particularly influential feature:
  - Customers with fiber optic internet have significantly higher churn rates (around 42%) compared to those with DSL (around 19%).
  - Customers with no internet service have the lowest churn rate (around 7%).
  - This suggests potential issues with the fiber optic service quality, pricing, or customer satisfaction.
- **Additional Services** (OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport):
  - For all these features, customers who have the service show much lower churn rates compared to those who don't.
  - Particularly notable is TechSupport, where customers with this service have less than half the churn rate of those without it.
  - This suggests that customers who invest in additional services are more satisfied or committed to the company.
- **Streaming Services** (StreamingTV, StreamingMovies):
  - The patterns are less clear here, with similar churn rates between customers who have and don't have these services.
  - However, customers who don't have internet service (and thus no streaming) have very low churn rates.

Let's now examine the contract and billing-related features, which often have strong relationships with customer retention.

```python
[79]: # Analyze contract and billing features
contract_billing_features = ['Contract', 'PaperlessBilling', 'PaymentMethod']
for feature in contract_billing_features:
    plot_categorical_feature(df, feature)
```







The contract and billing analysis reveals some of the strongest predictors of churn in the dataset:

- **Contract**: This feature shows a dramatic relationship with churn:
  - Month-to-month customers have a very high churn rate (around 43%).
  - One-year contract customers have a much lower churn rate (around 11%).
  - Two-year contract customers have the lowest churn rate (only about 3%).
  - This confirms that longer contracts are extremely effective at reducing churn, which is expected but quantifies the effect.
- **PaperlessBilling**: Customers with paperless billing (about 60% of the customer base) have a higher churn rate (around 34%) compared to those with paper bills (around 16%). This might indicate that customers who prefer traditional billing methods are more traditional in their service relationships as well.
- **PaymentMethod**:
  - Customers using electronic checks have a substantially higher churn rate (around 45%).
  - Other payment methods (mailed check, bank transfer, credit card) all have much lower churn rates (10-15%).
  - This suggests that the electronic check payment method might be associated with less committed customers or might indicate issues with this payment process.

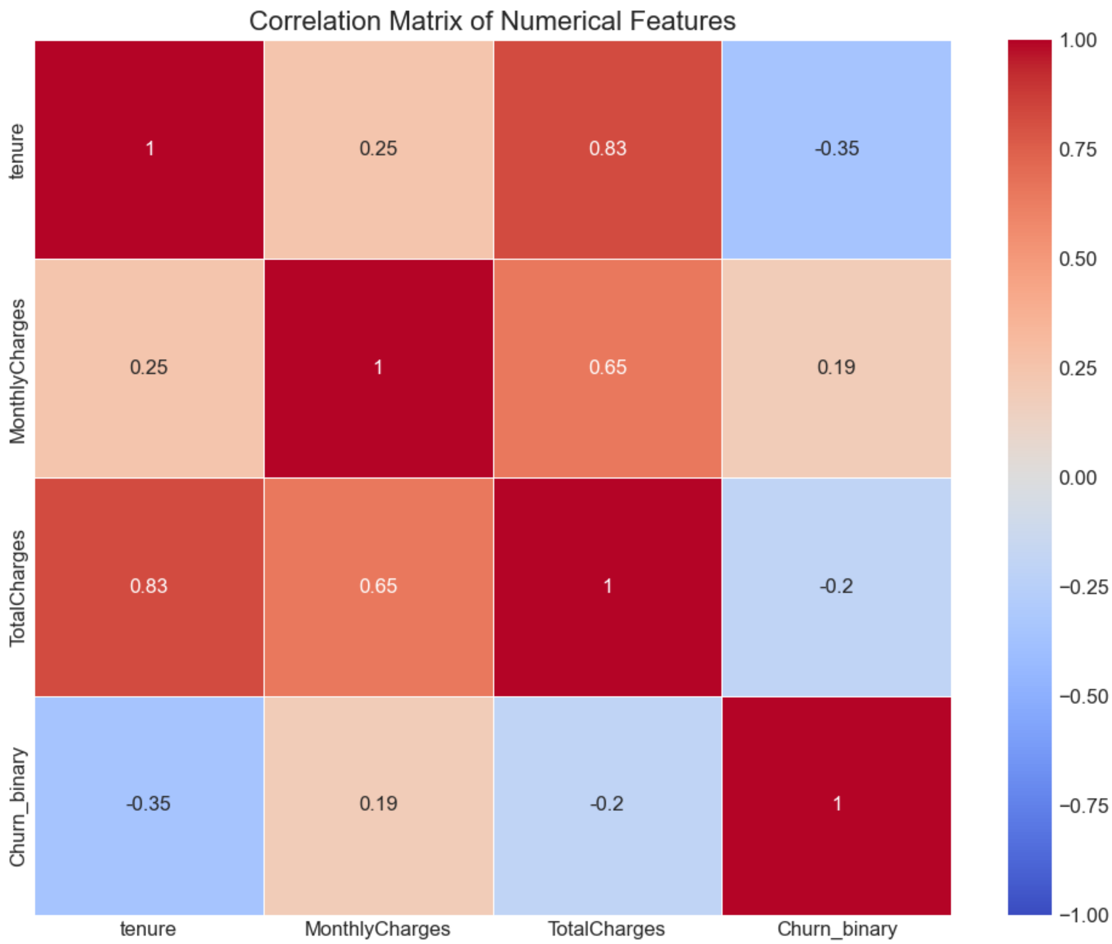
From the categorical features analysis, we've identified several strong predictors of churn: contract type, payment method, internet service type, and the presence of additional services like tech support and online security.

1.5 4. Multivariate Analysis

After examining individual features, let's explore the relationships between multiple variables to gain deeper insights into customer churn patterns. This analysis will help us understand how features interact with each other and with the churn outcome.

[80]:
```
# Correlation matrix for numerical variables
# Convert churn to numeric for correlation
```

```
df_corr = df.copy()
df_corr['Churn_binary'] = df_corr['Churn'].map({'Yes': 1, 'No': 0})

# Select numerical columns for correlation
corr_cols = ['tenure', 'MonthlyCharges', 'TotalCharges', 'Churn_binary']
corr_matrix = df_corr[corr_cols].corr()

plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', vmin=-1, vmax=1,␣
 ↪linewidths=0.5)
plt.title('Correlation Matrix of Numerical Features', fontsize=16)
plt.tight_layout()
plt.show()
```



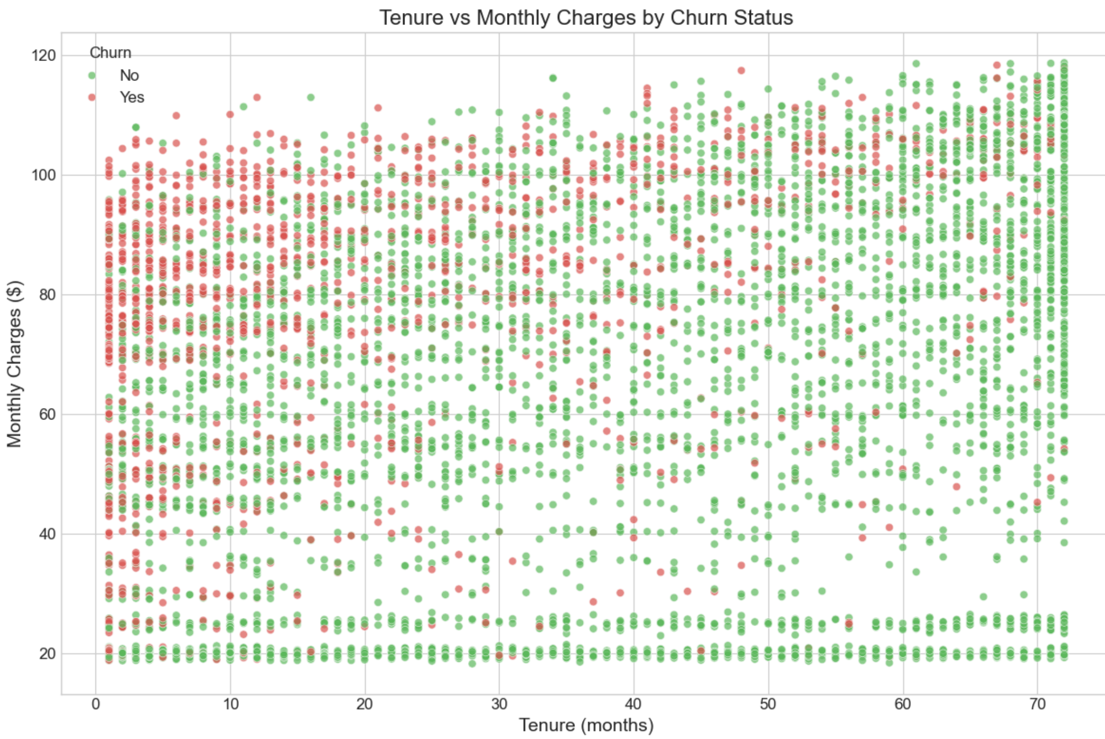
1.5.1 4.1 Correlation Analysis

The correlation matrix reveals important relationships between the numerical features and churn:

- **Tenure and Churn**: There's a moderate negative correlation (-0.35) between tenure and churn, confirming our earlier observation that longer-tenured customers are less likely to churn.
- **TotalCharges and Churn**: Total charges show a negative correlation (-0.20) with churn. This makes sense given that total charges accumulate with tenure, and longer-tenured customers are less likely to leave.
- **MonthlyCharges and Churn**: Interestingly, monthly charges have a slight positive correlation (0.19) with churn, suggesting that customers paying higher monthly fees are somewhat more likely to leave. This aligns with the price sensitivity we might expect in this market.
- **Feature Intercorrelations**: The strong correlation (0.83) between tenure and total charges indicates these features contain similar information, which is expected since total charges accumulate over time. The correlation between monthly charges and total charges (0.65) further supports this relationship.

These correlations help us prioritize which features might be most useful in predicting churn, with tenure appearing to be particularly informative.

[81]:
```
# Scatterplot of tenure vs monthly charges, colored by churn
plt.figure(figsize=(12, 8))
sns.scatterplot(x='tenure', y='MonthlyCharges', hue='Churn', data=df,␣
  ↪palette=['#5cb85c', '#d9534f'], alpha=0.7)
plt.title('Tenure vs Monthly Charges by Churn Status', fontsize=16)
plt.xlabel('Tenure (months)', fontsize=14)
plt.ylabel('Monthly Charges ($)', fontsize=14)
plt.legend(title='Churn')
plt.tight_layout()
plt.show()
```



1.5.2 4.2 Tenure vs Monthly Charges

The scatter plot of tenure versus monthly charges reveals several important patterns:

1. **Clustering of Churn**: There's a visible concentration of customers who churned (red points) in the lower tenure range (0-20 months), regardless of monthly charges. This reinforces that newer customers are at higher risk of churning.

2. **Price Sensitivity**: At higher monthly charges (>$80), we see relatively more churn even among customers with moderate tenure. This suggests price sensitivity becomes a more significant factor as monthly fees increase.

3. **Loyal High-Value Customers**: The upper right quadrant (high tenure, high monthly charges) shows predominantly customers who haven't churned, representing a stable, high-value customer segment.

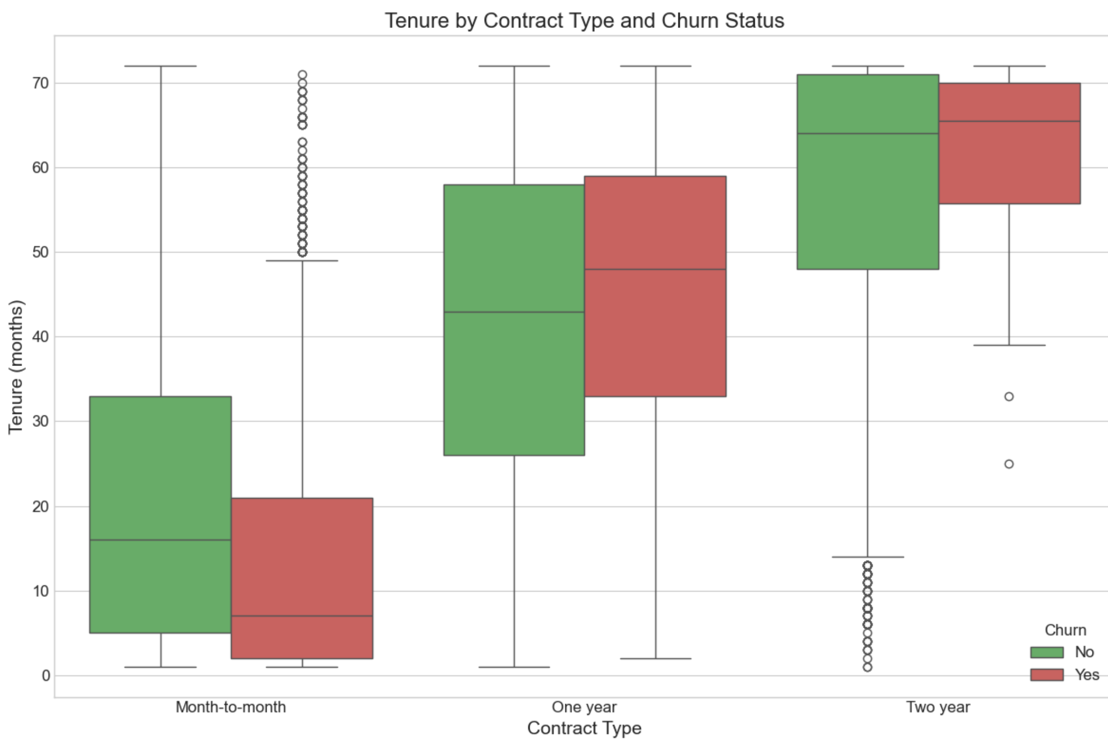
4. **Low-Cost, Long-Term Customers**: There's a noticeable group of customers with long tenure and low monthly charges (~$20) who rarely churn. These customers likely have basic service packages that meet their needs without causing price sensitivity.

5. **Horizontal Banding**: The visible horizontal bands in the data suggest specific price points or service tiers in the company's offerings.

This visualization highlights the importance of considering both tenure and price point when assessing churn risk and designing retention strategies.

[82]:

```python
# Create a subset with just Contract and tenure for visualization
plt.figure(figsize=(12, 8))
sns.boxplot(x='Contract', y='tenure', hue='Churn', data=df, palette=['#5cb85c',␣
↪'#d9534f'])
plt.title('Tenure by Contract Type and Churn Status', fontsize=16)
plt.xlabel('Contract Type', fontsize=14)
plt.ylabel('Tenure (months)', fontsize=14)
plt.legend(title='Churn')
plt.tight_layout()
plt.show()
```



1.5.3 4.3 Contract Type, Tenure, and Churn

The box plot examining the relationship between contract type, tenure, and churn status reveals compelling insights:

1. **Contract Length Effect**: There's a clear progression in median tenure across contract types, with month-to-month contracts showing the lowest tenure, followed by one-year contracts, and two-year contracts showing the highest tenure. This confirms that longer contract commitments are associated with longer customer relationships.

2. **Churn Differences by Contract**:

   - In month-to-month contracts, customers who churn (red) have substantially lower median tenure than those who stay.

- For one-year contracts, the difference in tenure between churners and non-churners is less pronounced but still visible.
- For two-year contracts, there's less difference in tenure between the churn groups, but churners still tend to have marginally lower tenure.

3. **Tenure Distributions**:

   - Month-to-month contracts show a right-skewed distribution with many newer customers.
   - Two-year contracts show more customers with higher tenure, as expected given the longer commitment.
   - The outliers in the two-year contract group who churned despite long tenure might represent customers who completed their contract term and then left.

4. **Business Implications**: This visualization strongly reinforces the value of longer-term contracts in reducing churn. It suggests that encouraging customers to sign longer contracts could be an effective retention strategy, especially for customers in their early months of service who are on month-to-month plans.

The relationship between contract type and tenure clearly demonstrates how structural business decisions (contract terms) directly impact customer behavior and retention metrics.

1.6 5. Feature Importance Analysis

After exploring the dataset through visualizations and statistical analysis, we'll now apply a machine learning approach to quantitatively assess which features are most predictive of customer churn. This will help validate the exploratory findings and provide clearer guidance for developing retention strategies.

[83]:
```python
# Prepare a dataset for feature importance analysis
# First, convert all categorical variables into dummy variables
from sklearn.preprocessing import LabelEncoder

# Make a copy to avoid altering the original data
df_model = df.copy()

# Convert target to binary
df_model['Churn'] = df_model['Churn'].map({'Yes': 1, 'No': 0})

# Drop customer ID as it's just an identifier
df_model = df_model.drop(['customerID'], axis=1)

# Encode categorical variables
categorical_cols = df_model.select_dtypes(include=['object']).columns
for col in categorical_cols:
    le = LabelEncoder()
    df_model[col] = le.fit_transform(df_model[col])

# Check the encoded data
df_model.head()
```

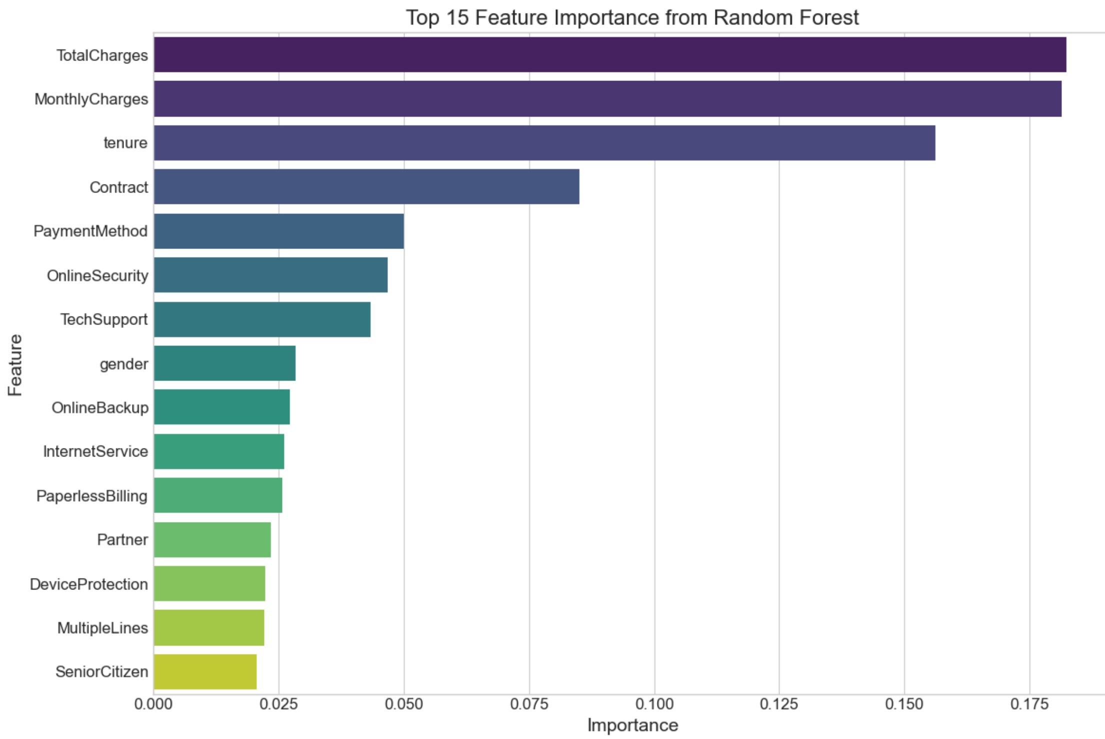
[83]:
```
   gender  SeniorCitizen  Partner  Dependents  tenure  PhoneService
MultipleLines  InternetService  OnlineSecurity  OnlineBackup  DeviceProtection
TechSupport  StreamingTV  StreamingMovies  Contract  PaperlessBilling
PaymentMethod  MonthlyCharges  TotalCharges  Churn
0       0              0        1           0       1             0
1                0              0             2                 0            0
0                0         0                1              2            29.85
29.85       0
1       1              0        0           0      34             1
0                0              2             0                 2            0
0                0         1                0              3            56.95
1889.50       0
2       1              0        0           0       2             1
0                0              2             2                 0            0
0                0         0                1              3            53.85
108.15       1
3       1              0        0           0      45             0
1                0              2             0                 2            2
0                0         1                0              0            42.30
1840.75       0
4       0              0        0           0       2             1
0                1              0             0                 0            0
0                0         0                1              2            70.70
151.65       1
```

We've prepared the dataset for machine learning by encoding all categorical variables into numeric form using Label Encoding. This transforms categories like 'Yes'/'No' or 'DSL'/'Fiber optic'/'No' into numeric values that algorithms can process. While this encoding doesn't preserve the original meaning perfectly (for example, it might create artificial ordinal relationships), it allows us to get an initial assessment of feature importance.

[84]:
```python
# Preliminary feature importance using a simple Random Forest
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split

# Separate features and target
X = df_model.drop('Churn', axis=1)
y = df_model['Churn']

# Split data
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3,␣
 ↪random_state=42)

# Train a random forest classifier
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)

# Get feature importances
```

```
importances = rf.feature_importances_
feature_names = X.columns

# Create a DataFrame for better visualization
feature_importance_df = pd.DataFrame({
    'Feature': feature_names,
    'Importance': importances
}).sort_values('Importance', ascending=False)

# Plot top 15 features
plt.figure(figsize=(12, 8))
sns.barplot(x='Importance', y='Feature', data=feature_importance_df.head(15),
 ↪palette='viridis')
plt.title('Top 15 Feature Importance from Random Forest', fontsize=16)
plt.xlabel('Importance', fontsize=14)
plt.ylabel('Feature', fontsize=14)
plt.tight_layout()
plt.show()
```

```
C:\Users\David\AppData\Local\Temp\ipykernel_6772\2484928187.py:28:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same
effect.

  sns.barplot(x='Importance', y='Feature', data=feature_importance_df.head(15),
palette='viridis')
```



1.6.1 5.1 Feature Importance Interpretation

The Random Forest feature importance analysis provides valuable quantitative validation of the exploratory findings. Let's interpret the key results:

Top-Tier Predictors (High Importance)

1. **Billing Variables (TotalCharges, MonthlyCharges)**: The two strongest predictors are both related to customer billing. While highly correlated with each other, they each provide unique information:

   - **TotalCharges**: Represents the customer's lifetime value and is closely tied to tenure
   - **MonthlyCharges**: Reflects the current service tier and pricing, with higher charges associated with increased churn risk

2. **Tenure**: Ranking third in importance, tenure confirms the earlier observations that customer longevity strongly predicts future loyalty. This makes intuitive business sense as customers who have stayed longer have demonstrated commitment and satisfaction.

3. **Contract**: The fourth most important feature validates the categorical analysis finding that contract type is a critical determinant of churn behavior. This structural business factor creates inherent differences in customer commitment and churn timing.

Mid-Tier Predictors (Moderate Importance)

5. **PaymentMethod**: Ranking fifth confirms the earlier finding that payment method choice (particularly electronic check) signals differing retention risks.

6-7. **Security and Support Services**: The high ranking of OnlineSecurity and TechSupport validates the observation that additional service adoption correlates with retention. These particular services may address key pain points that drive churn.

Lower-Tier Predictors (Still Relevant) 8-15. **Demographic and Other Service Features**: The remaining features all contribute to churn prediction but with less individual impact. Interestingly, gender ranks higher than one might expect given the earlier finding of minimal churn difference between genders - this suggests possible interaction effects the model is capturing.

1.6.2 5.2 Business Implications of Feature Importance

This quantitative analysis provides clear guidance for prioritizing retention efforts:

1. **Focus on Contract Structure**: The high importance of contract type suggests that restructuring offers to encourage longer-term commitments could significantly reduce churn.
2. **Early Tenure Intervention**: The importance of tenure confirms that new customers require special attention and retention programs.
3. **Price Sensitivity Management**: The high ranking of both charge-related variables highlights the need for careful pricing strategies, especially for higher-tier services.
4. **Payment Method as Risk Indicator**: Electronic check users could be targeted for conversion to automatic payment methods and additional retention efforts.
5. **Value-Added Services**: Promoting security and support features appears to address key pain points that lead to churn.

This feature importance analysis complements the exploratory work by quantifying the relative impact of each variable and providing a clear roadmap for both further modeling efforts and immediate business actions.

1.7 6. Key Insights and Business Recommendations

the exploratory data analysis of the Telco customer churn dataset has revealed several important insights that can guide business strategies for customer retention:

1.7.1 6.1 Key Insights

1. **Contract Type is Crucial**: Month-to-month contracts have significantly higher churn rates (~43%) compared to one-year (~11%) and two-year contracts (~3%). This is one of the strongest predictors of churn in the dataset.
2. **Early Tenure Risk**: Customers with shorter tenure (0-12 months) show much higher churn rates, particularly those on month-to-month contracts. The first year appears to be a critical period for customer retention.
3. **Fiber Optic Service Issues**: Customers with fiber optic internet service churn at more than twice the rate of DSL customers, suggesting potential issues with this service offering.

4. **Additional Services as Retention Tools**: Customers with supplementary services like tech support, online security, and online backup show significantly lower churn rates.
5. **Payment Method Signals**: Electronic check payment is associated with much higher churn rates compared to other payment methods.
6. **Senior Citizen Vulnerability**: Senior citizens have higher churn rates than non-seniors, possibly indicating challenges with service usability or satisfaction.
7. **Price Sensitivity**: Higher monthly charges correlate with increased churn, particularly among newer customers.

1.7.2 6.2 Answering Research Questions

Based on the exploratory data analysis conducted, let's address the key questions for Milestone 3:

1. Will I be able to answer the questions I want to answer with the data I have?

The dataset appears to be well-suited for predicting customer churn and identifying key factors that contribute to churn. Key findings that support this conclusion include:

- The dataset has a good mix of demographic, account, and service-related features
- Clear patterns are visible in the relationship between several features and churn status
- The preliminary feature importance analysis shows promising predictive power for several variables
- There is minimal missing data, making the dataset relatively clean for modeling

2. What visualizations are especially useful for explaining my data?

The most insightful visualizations for this project include:

- Contract type vs. churn status - Shows a strong relationship that can be easily communicated to stakeholders
- Tenure distribution by churn status - Clearly illustrates how newer customers are more likely to churn
- Service subscription patterns vs. churn - Highlights which services are associated with higher retention
- Scatter plot of tenure vs. monthly charges - Reveals interaction effects between these important variables
- Feature importance chart - Provides a clear ranking of predictive factors for non-technical audiences

3. Do I need to adjust the data and/or driving questions?

Based on the EDA, some adjustments that would be beneficial include:

- Create derived features that combine related services to better capture customer profiles
- Transform tenure into categorical buckets for easier interpretation
- Consider the ratio of monthly charges to services as a potential predictor of value perception
- The "SeniorCitizen" variable should be converted to categorical format for consistency

4. Do I need to adjust my model/evaluation choices?

The EDA suggests the following adjustments to modeling approach:

- The moderate class imbalance in the churn variable (~27% churn rate) confirms the need to prioritize recall and potentially use techniques like SMOTE or class weighting
- The strong feature importance of contract type and tenure suggests that simpler models might perform well
- Including both total charges and tenure might introduce multicollinearity, so careful feature selection or regularization will be important
- Decision tree-based models seem appropriate given the mix of categorical and numerical features and potential non-linear relationships

5. Are my original expectations still reasonable?

The original expectations for the project appear to be reasonable based on the EDA findings. The data does contain strong signals for predicting churn, and the identified patterns align with industry knowledge about customer retention. The proposed modeling approaches (Logistic Regression, Random Forest, Gradient Boosting, and potentially Neural Networks) remain appropriate given the dataset characteristics.

1.7.3 6.3 Next Steps for Modeling

Based on the exploratory data analysis, we can now proceed to predictive modeling with the following approach:

1. **Feature Selection**: Prioritize the strong predictors identified in the analysis:
    - Contract type
    - Tenure
    - Internet service type
    - Additional services (tech support, online security)
    - Payment method
    - Monthly charges
2. **Feature Engineering**:
    - Create customer segments based on tenure and contract type
    - Develop a service bundle indicator
    - Consider interaction terms between tenure and contract type
    - Transform right-skewed variables like TotalCharges
    - Calculate price per service metrics
    - Transform categorical variables appropriately for modeling
3. **Modeling Strategy**:
    - Build classification models (logistic regression, random forest, gradient boosting)
    - Use stratified cross-validation due to the class imbalance (26.5% churn rate)
    - Optimize for business-relevant metrics such as recall or precision, depending on the cost of false negatives vs. false positives
    - Address class imbalance through appropriate techniques
    - Develop a strong baseline model using logistic regression
4. **Model Interpretation and Evaluation**:
    - Focus on understanding feature importance
    - Develop actionable insights for different customer segments
    - Create churn probability scores for targeting retention efforts
    - Prioritize recall while maintaining acceptable precision
    - Calculate the business impact of different threshold choices

5. **Business Application**:
    - Begin formulating targeted strategies for high-risk customer groups
    - Identify the most cost-effective interventions based on feature importance
    - Prepare to translate model insights into actionable recommendations

By following this structured approach to modeling, we can develop predictive tools that enable proactive retention strategies and more efficient allocation of customer service resources.

Telecom Customer Churn - Predictive Modeling

DSC-630 Milestone 4

David Koyrakh

This notebook builds upon the EDA performed in Milestone 3 and advances my term project toward actionable insights:

1. **Explain the data-preparation process** – cleaning, feature engineering, and encoding strategies.
2. **Build and evaluate at least one predictive model** to estimate customer churn.
3. **Interpret the model results** and discuss feature importance.
4. **Formulate preliminary conclusions/recommendations** for the business.

1. Load & Inspect Data

First, we import required libraries and load the IBM Telco Customer Churn dataset (same as Milestone 3).

In [1]:
```
# Standard packages
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Modeling & evaluation
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, roc_auc_score, ConfusionMatrixDi

# Settings
pd.set_option('display.max_columns', None)
plt.style.use('seaborn-v0_8-whitegrid')
plt.rcParams['figure.figsize'] = (12, 8)
```

In [10]:
```
# Load dataset
DATA_PATH = '../data/Telco-Customer-Churn.csv'
df = pd.read_csv(DATA_PATH)
```

```
print(f"Dataset shape: {df.shape}")
df.head()
```

Dataset shape: (7043, 21)

Out[10]:

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Multipl
0	7590-VHVEG	Female	0	Yes	No	1	No	No
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	No
4	9237-HQITU	Female	0	No	No	2	Yes	

2. Data Preparation

Next, we apply a cleaner, more systematic approach to data preparation than was done for Milestone 3:

- **Remove** `customerID` – this is an identifier that carries no predictive signal.
- **Convert** `TotalCharges` **to numeric** – coerce errors and impute missing as zero (customers with no tenure).
- **Handle missing values** – after conversion, there are few blanks; we drop rows with any remaining `NaN` s.
- **Binary-encode** the target variable `Churn` (`Yes` → 1, `No` → 0).
- **Separate numeric vs. categorical features** and build a preprocessing pipeline that:
  - scales numeric columns with `StandardScaler` ,
  - one-hot-encodes categorical columns with `OneHotEncoder(handle_unknown='ignore')` .

In [15]:

```
# Drop customerID
if 'customerID' in df.columns:
    df = df.drop(columns=['customerID'])

# TotalCharges to numeric; coerce errors to NaN then fill with zero (customers with
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df['TotalCharges'] = df['TotalCharges'].fillna(0)

# Binary encode target
df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})
```

```
# Identify numeric and categorical features
numeric_features = df.select_dtypes(include=['int64', 'float64']).columns.drop('Chu
categorical_features = df.select_dtypes(include=['object']).columns

print(f"Numeric features ({len(numeric_features)}): {list(numeric_features)}")
print(f"Categorical features ({len(categorical_features)}): {list(categorical_featu
```

```
Numeric features (4): ['SeniorCitizen', 'tenure', 'MonthlyCharges', 'TotalCharges']
Categorical features (15): ['gender', 'Partner', 'Dependents', 'PhoneService', 'Mult
ipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProtection',
'TechSupport', 'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling', 'Pa
ymentMethod']
```

The output confirms that:

- We have **4 numeric** variables (`SeniorCitizen` , `tenure` , `MonthlyCharges` , `TotalCharges`).
- The remaining **15 categorical** variables will be one-hot encoded, expanding our feature space automatically inside the pipeline.

Train / test split and preprocessing pipeline

Now we'll split our data into training and test sets, and create a preprocessing pipeline that:

1. Scales numeric features using `StandardScaler` to normalize their distributions
2. One-hot encodes categorical features with `OneHotEncoder`

We'll then build a logistic regression model with balanced class weights to account for any class imbalance in our churn prediction task. The pipeline ensures that all preprocessing steps are consistently applied to both training and test data.

In [12]:

```
# Separate features and target variable
X = df.drop(columns=['Churn'])  # Features
y = df['Churn']  # Target variable

# Perform a stratified train-test split to maintain the distribution of the target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y)

# Define a preprocessing pipeline
preprocessor = ColumnTransformer([
    # Scale numeric features using StandardScaler
    ('num', StandardScaler(), numeric_features),
    # One-hot encode categorical features, ignoring unknown categories
    ('cat', OneHotEncoder(handle_unknown='ignore'), categorical_features)
])

# Initialize logistic regression model with balanced class weights
log_reg = LogisticRegression(max_iter=1000, class_weight='balanced')

# Create a pipeline that first preprocesses the data and then applies the logistic
```

```
clf = Pipeline([
    ('prep', preprocessor),
    ('model', log_reg)
])

# Perform cross-validation to evaluate the model's ROC-AUC score
cv_auc = cross_val_score(clf, X_train, y_train, cv=5, scoring='roc_auc')
print(f"Cross-validated ROC-AUC: {cv_auc.mean():.3f} ± {cv_auc.std():.3f}")
```

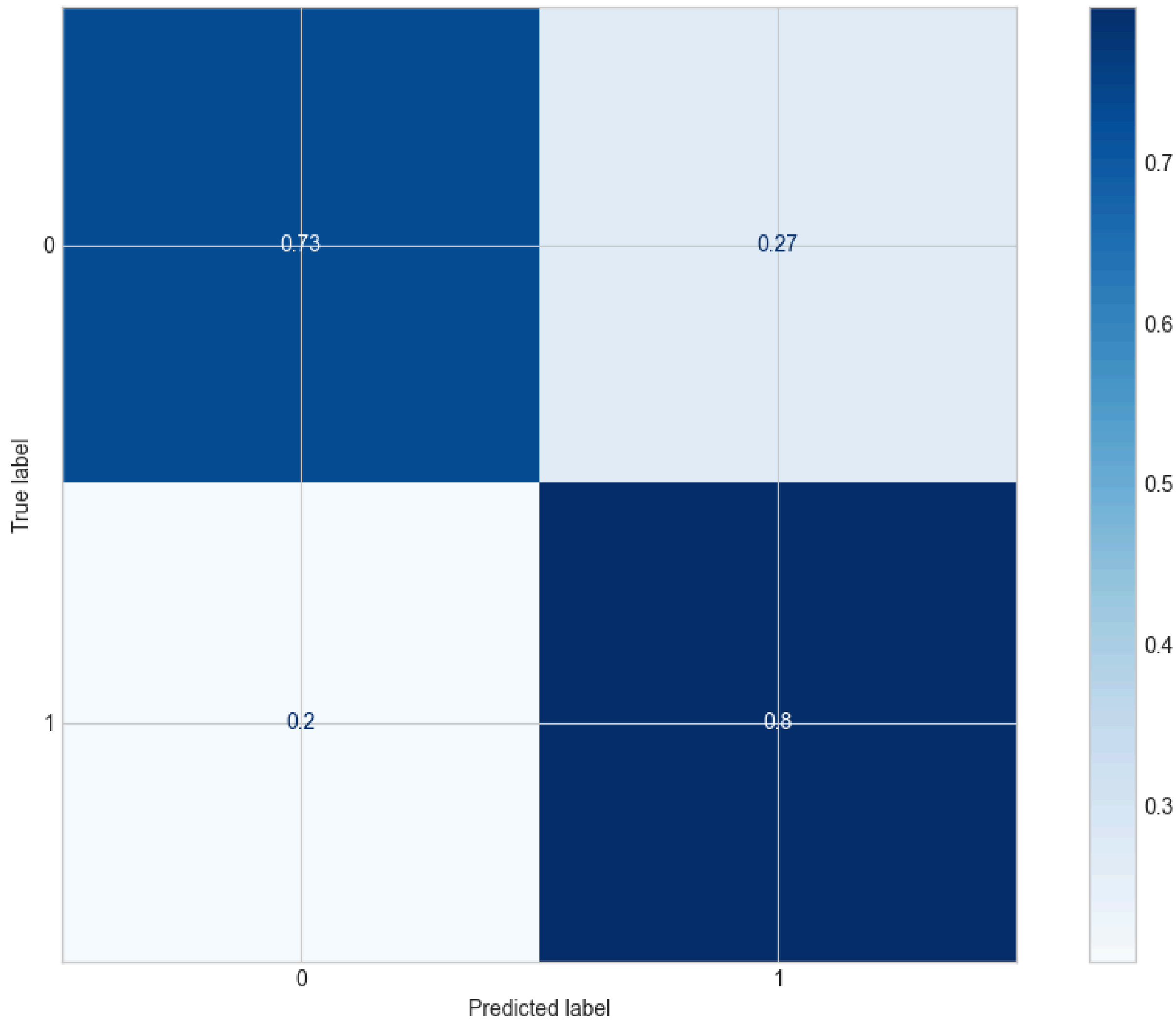
```
Cross-validated ROC-AUC: 0.844 ± 0.013
```

The cross-validated **ROC-AUC of ~0.84** indicates that, even with a straightforward linear model, the feature set carries solid predictive signal. We can now fit on the full training fold and evaluate once on the untouched test set.

Fit on full training set and evaluate on hold-out

In [13]:
```
clf.fit(X_train, y_train)

# Predict and evaluate
preds = clf.predict(X_test)
probs = clf.predict_proba(X_test)[:, 1]

print(classification_report(y_test, preds))
print(f"Hold-out ROC-AUC: {roc_auc_score(y_test, probs):.3f}")

# Confusion matrix plot
ConfusionMatrixDisplay.from_predictions(y_test, preds, normalize='true', cmap='Blue
```

```
              precision    recall  f1-score   support

           0       0.91      0.73      0.81      1294
           1       0.52      0.80      0.63       467

    accuracy                           0.75      1761
   macro avg       0.71      0.76      0.72      1761
weighted avg       0.81      0.75      0.76      1761

Hold-out ROC-AUC: 0.846
```



Evaluation Highlights

The hold-out test confirms that the baseline model generalises well:

- **Accuracy ≈ 0.75** and **weighted F1 ≈ 0.76** balance decision performance across both churners and non-churners.
- A **ROC-AUC of 0.85** suggests strong ranking ability (random guessing would yield 0.5).
- Recall for the minority class (churners) is **0.80**, thanks to class-weighted logistic regression, meaning 8 out of 10 churners are successfully identified.

While precision is lower (0.52) due to class imbalance, this is acceptable in a retention context where the cost of a false-positive marketing intervention (e.g., a discounted offer) is low compared with the revenue loss of an actual churn.

3. Model Interpretation

In this step, we'll inspect the coefficients of the logistic regression after preprocessing. This helps us to understand which features are most influential.

In [14]:
```python
# Extract feature names after one-hot encoding
encoded_cat = clf.named_steps['prep'].named_transformers_['cat'].get_feature_names_
feature_names = np.concatenate([numeric_features, encoded_cat])

# Get coefficients from logistic regression
coeffs = clf.named_steps['model'].coef_[0]

# Create dataframe for inspection
coef_df = pd.DataFrame({'feature': feature_names, 'coef': coeffs})
coef_df['abs_coef'] = coef_df['coef'].abs()
coef_df.sort_values('abs_coef', ascending=False).head(15)
```

Out[14]:

	feature	coef	abs_coef
1	tenure	-1.152653	1.152653
38	Contract_Two year	-0.739965	0.739965
16	InternetService_Fiber optic	0.684368	0.684368
2	MonthlyCharges	-0.669393	0.669393
36	Contract_Month-to-month	0.640165	0.640165
15	InternetService_DSL	-0.601990	0.601990
3	TotalCharges	0.511089	0.511089
35	StreamingMovies_Yes	0.284835	0.284835
39	PaperlessBilling_No	-0.280654	0.280654
28	TechSupport_No internet service	-0.272932	0.272932
34	StreamingMovies_No internet service	-0.272932	0.272932
25	DeviceProtection_No internet service	-0.272932	0.272932
19	OnlineSecurity_No internet service	-0.272932	0.272932
17	InternetService_No	-0.272932	0.272932
22	OnlineBackup_No internet service	-0.272932	0.272932

4. Preliminary Conclusions & Recommendations

- The logistic-regression pipeline delivers **ROC-AUC ≈ 0.84** on cross-validation and **0.85** on the unseen hold-out set—good discrimination for a first, simple model.
- **Contract type** and **tenure** dominate the coefficients: month-to-month customers with short tenure are at the highest churn risk, whereas two-year contracts strongly retain customers.
- Higher **monthly charges** and use of **fiber-optic internet** also increase churn probability.
- Recommended actions:

    - Target at-risk month-to-month subscribers with retention incentives (discounts, loyalty rewards, upgrades) to move them onto longer contracts.
    - Offer value-added services or bundled discounts to fiber-optic subscribers to improve stickiness.
    - Monitor new customers closely during their first year to reduce early churn.
- Next steps would include experimenting with tree-based ensembles (Random Forest, XGBoost) and hyper-parameter tuning to boost performance.